Docker Network :

Creating 2 Containers

```
ip-172-31-10-73:/ # docker run -itd --name c1 alpine
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
e2de96513ba9: Pull complete
6d0606d1815c: Download complete
797dd00a0fc7: Download complete
Digest: sha256:294b683cb724975bec92580e1e685676bd4b50bda910ddb8c51d4cabeaec77e6
Status: Downloaded newer image for alpine:latest
9ff07f74e396ba2e0f1c440f1cd2e6ac5c77ffc02955a6de15e2b13599203ec6
ip-172-31-10-73:/ # docker run -itd --name c2 alpine
8959fe40afd96f094c0570044f86cadf714f879cd1bab4a4ed5d21e7853539de
```

```
ip-172-31-10-73:/ # docker ps
CONTAINER ID   IMAGE     COMMAND     CREATED          STATUS          PORTS
NAMES
8959fe40afd9   alpine    "/bin/sh"   34 seconds ago   Up 33 seconds
c2
9ff07f74e396   alpine    "/bin/sh"   42 seconds ago   Up 42 seconds
c1
```

```
ip-172-31-10-73:/ # docker inspect c2
[
    {
        "Id": "8959fe40afd96f094c0570044f86cadf714f879cd1bab4a4ed5d21e7853539de"
,
        "Created": "2026-10-17T18:48:07.807463447Z",
        "Path": "/bin/sh",
        "Args": [],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 9034,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-10-17T18:48:07.958351041Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:294b683cb724975bec92580e1e685676bd4b50bda910ddb8c51d4ca
beaec77e6",
        "ResolvConfPath": "/var/lib/docker/containers/8959fe40afd96f094c0570044f
86cadf714f879cd1bab4a4ed5d21e7853539de/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/8959fe40afd96f094c0570044f86
cadf714f879cd1bab4a4ed5d21e7853539de/hostname",
        "HostsPath": "/var/lib/docker/containers/8959fe40afd96f094c0570044f86cad
f714f879cd1bab4a4ed5d21e7853539de/hosts",
        "LogPath": "/var/lib/docker/containers/8959fe40afd96f094c0570044f86cadf7
14f879cd1bab4a4ed5d21e7853539de/8959fe40afd96f094c0570044f86cadf714f879cd1bab4a4
ed5d21e7853539de-json.log",
        "Name": "/c2",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "system_u:object_r:container_file_t:s0:c221,c804",
        "ProcessLabel": "system_u:system_r:container_t:s0:c221,c804",
        "AppArmorProfile": "",
        "ExecIDs": null
```

```
        },
        "NetworkSettings": {
            "SandboxID": "abdcbfd4ed4811ee977ebd0c6a1b334813ca028743a445f6a27788
324804e710",
            "SandboxKey": "/var/run/docker/netns/abdcbfd4ed48",
            "Ports": {},
            "Networks": {
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "DriverOpts": null,
                    "GwPriority": 0,
                    "NetworkID": "486ab2c524f7a009715b337160d437dfe3cdbc6f48a20d
418b1367935e56c18e",
                    "EndpointID": "fb148ab7f4dcad4450c8087c9df24161ba5dccbdd13a7
a02f9d9117e40f3c506",
                    "Gateway": "172.17.0.1",
                    "IPAddress": "172.17.0.3",
                    "MacAddress": "e6:56:c8:24:ff:83",
                    "IPPrefixLen": 16,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DNSNames": null
                }
            }
        }
```

Copy the ip Address

Ping From the first Container Using the Ip :

```
ip-172-31-10-73:/ # docker exec -it c1 sh
/ # ping 172.17.0.3
PING 172.17.0.3 (172.17.0.3): 56 data bytes
64 bytes from 172.17.0.3: seq=0 ttl=64 time=0.099 ms
64 bytes from 172.17.0.3: seq=1 ttl=64 time=0.077 ms
64 bytes from 172.17.0.3: seq=2 ttl=64 time=0.074 ms
64 bytes from 172.17.0.3: seq=3 ttl=64 time=0.082 ms
64 bytes from 172.17.0.3: seq=4 ttl=64 time=0.072 ms
64 bytes from 172.17.0.3: seq=5 ttl=64 time=0.074 ms
64 bytes from 172.17.0.3: seq=6 ttl=64 time=0.071 ms
64 bytes from 172.17.0.3: seq=7 ttl=64 time=0.077 ms
64 bytes from 172.17.0.3: seq=8 ttl=64 time=0.075 ms
64 bytes from 172.17.0.3: seq=9 ttl=64 time=0.101 ms
64 bytes from 172.17.0.3: seq=10 ttl=64 time=0.070 ms
64 bytes from 172.17.0.3: seq=11 ttl=64 time=0.074 ms
64 bytes from 172.17.0.3: seq=12 ttl=64 time=0.124 ms
64 bytes from 172.17.0.3: seq=13 ttl=64 time=0.072 ms
```

Creating the Own Network :

```
ip-172-31-10-73:/ # docker network create --driver bridge uday-net
c9b02aeccfb7e5b06887230171187ddd3134f419fa6371a831ce9fab7daaec34
```

Creating the Containers in the Network :

```
ip-172-31-10-73:/ # docker run -itd --name ne1 --network uday-net alpine
1059fdba2c1ec6048ee43477f3cfc0ee854e01530824d624c7d37b08e5d83458
ip-172-31-10-73:/ # docker run -itd --name ne2 --network uday-net alpine
6de0519bbb4fbdd7b21b50b2b3628348d19740c48815adb43c9da2f7d5c98ff6
ip-172-31-10-73:/ # docker ps
CONTAINER ID   IMAGE     COMMAND     CREATED          STATUS          PORTS     NAMES
6de0519bbb4f   alpine    "/bin/sh"   12 seconds ago   Up 12 seconds             ne2
1059fdba2c1e   alpine    "/bin/sh"   19 seconds ago   Up 18 seconds             ne1
8959fe40afd9   alpine    "/bin/sh"   6 minutes ago    Up 6 minutes              c2
9ff07f74e396   alpine    "/bin/sh"   6 minutes ago    Up 6 minutes              c1
```

Pinging the ne2 from ne1 by name :

```
ip-172-31-10-73:/ # docker exec -it ne1 sh
/ # ping ne2
PING ne2 (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.083 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.081 ms
64 bytes from 172.18.0.3: seq=2 ttl=64 time=0.079 ms
^C
--- ne2 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.079/0.081/0.083 ms
/ #
```